

Prüfbericht-Nr.: <i>Test report no.:</i>	50183074 003	Auftrags-Nr.: <i>Order no.:</i>	168476022	Seite 1 von 5 Page 1 of 5
Kunden-Referenz-Nr.: <i>Client reference no.:</i>	N/A	Auftragsdatum: <i>Order date:</i>	2024.03.26	
Auftraggeber: <i>Client:</i>	Xinxiang Hongli Supply Source Technology Co.,Ltd. West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China			
Prüfgegenstand: <i>Test item:</i>	CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL			
Bezeichnung / Typ-Nr.: <i>Identification / Type no.:</i>	IMR18650-1500mAh, IMR18650-1200mAh			
Auftrags-Inhalt: <i>Order content:</i>	TÜVus mark approval			
Prüfgrundlage: <i>Test specification:</i>	UL 1642:2020 R10.22			
Wareneingangsdatum: <i>Date of sample receipt:</i>	N/A	See attachment 1 for the photo documentation		
Prüfmuster-Nr.: <i>Test sample no.:</i>	N/A			
Prüfzeitraum: <i>Testing period:</i>	N/A			
Ort der Prüfung: <i>Place of testing:</i>	Guangzhou MCM Certification & Testing Co., Ltd.			
Prüflaboratorium: <i>Testing laboratory:</i>	TÜV Rheinland (Shenzhen) Co., Ltd.			
Prüfergebnis*: <i>Test result*:</i>	Pass			
erstellt von: <i>created by:</i>		genehmigt von: <i>authorized by:</i>		
Datum: <i>Date:</i>	2024.04.24	Ausstellungsdatum: <i>Issue date:</i>	2024.04.24	
Stellung / Position:	Project Engineer	Stellung / Position:	Reviewer	
Sonstiges / <i>Other:</i>	The complete test report includes the following attachments: -Attachment 1: Photo documentation (2 pages)			
Zustand des Prüfgegenstandes bei Anlieferung: <i>Condition of the test item at delivery:</i>	Prüfmuster vollständig und unbeschädigt <i>Test item complete and undamaged</i>			
* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
* Legend:	P(ass) = passed a.m. test specification(s)	F(ail) = failed a.m. test specification(s)	N/A = not applicable	N/T = not tested
Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens. <i>This test report only relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.</i>				

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Test report no.:

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Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. <i>The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.</i> <i>Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.</i>
2	Wie vertraglich vereinbart, wurde dieses Dokument nur digital unterzeichnet. Der TÜV Rheinland hat nicht überprüft, welche rechtlichen oder sonstigen diesbezüglichen Anforderungen für dieses Dokument gelten. Diese Überprüfung liegt in der Verantwortung des Benutzers dieses Dokuments. Auf Verlangen des Kunden kann der TÜV Rheinland die Gültigkeit der digitalen Signatur durch ein gesondertes Dokument bestätigen. Diese Anfrage ist an unseren Vertrieb zu richten. Eine Umweltgebühr für einen solchen zusätzlichen Service wird erhoben. Informationen zur Verifizierung der Authentizität unserer Dokumente erhalten Sie über folgenden Link: Einführung in digitale Signaturen <i>As contractually agreed, this document has been signed digitally only. TÜV Rheinland has not verified and unable to verify which legal or other pertaining requirements are applicable for this document. Such verification is within the responsibility of the user of this document. Upon request by its client, TÜV Rheinland can confirm the validity of the digital signature by a separate document. Such request shall be addressed to our Sales department. An environmental fee for such additional service will be charged. For information on verifying the authenticity of our documents, please visit the following link: Introduction to Digital Signature</i>
3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. <i>Test clauses with remark of * are subcontracted to qualified subcontractors and described under the respective test clause in the report.</i> <i>Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.</i>
4	Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnissen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezüglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019. <i>The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the "Zero Guard Band Rule" and "Simple Acceptance" in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.</i>

Test item particulars:

Information about the product needed to establish a correct test program, such as product mobility, type of power connections and similar.	(Test item particulars are selected by the TRF Originator base on the requirements in the standard)
Designation.....	: IMR18650-1500mAh, IMR18650-1200mAh
Trademark	: N/A
Nominal voltage.....	: DC 3.7V
Rated capacity.....	: 1500mAh (for IMR18650-1500mAh), 1200mAh (for IMR18650-1200mAh)
Upper limit charging voltage.....	: 4.2V
Maximum charge current.....	: 750mA (for IMR18650-1500mAh), 600mA (for IMR18650-1200mAh)
Manufacturer's specified end point voltage.....	: 3.0V
Operating temperature range	: 0~45°C (charge), -10~60°C (discharge)
Recommend charging method declared by the manufacturer	: Charge the cell with 0.2C constant current to 4.2V, then 4.2V constant voltage until charging current reaches 0.02C at ambient 20°C±5°C.
Utilization Type	: Technician replaceable cell

Testing:
Test clauses and results:

cl.10 Short-Circuit Test.....	: N/A
cl.11 Abnormal Charging Test.....	: N/A
cl.12 Forced-Discharge Test	: N/A
cl.13 Crush Test	: N/A
cl.14 Impact Test	: N/A
cl.14A Round Bar Crush Test.....	: N/A
cl.15 Shock Test.....	: N/A
cl.16 Vibration Test.....	: N/A
cl.17 Heating Test	: N/A
cl.18 Temperature Cycling Test	: N/A
cl.19 Low Pressure (Altitude Simulation) Test	: N/A
cl.20 Projectile Test.....	: N/A

Testing location:

Guangzhou MCM Certification & Testing Co., Ltd.
Room 101 to 116 & 216, Building 2 (Office Building and Workshop)No. 45 Zhong Er Section of Shiguang Road, Zhongcun Street, Panyu District, Guangzhou City, Guangdong Province, China

General remarks:

This report shall not be reproduced, except in full, without the written approval of the testing laboratory.
The test results presented in this report relate only to the object tested.
"(see remark #)" refers to a remark appended to the report.
"(see appended table)" refers to a table appended to the report.
Throughout this report a point is used as the decimal separator.

Copy of marking plate:

See original report 50183074 001and 50183074 002.

General product information:

This cell consists of the positive electrode plate, negative electrode plate, separator and electrolyte. The positive and negative electrode plates are housed in the case in the state being separated by the separator.

The cell is evaluated according to **UL 1642:2020 R10.22** in this test report, and passed all single-cell tests.

1) These tested cells have not been evaluated in combination with charger(s) or host product(s). Additional evaluation to determine compliance may be required on the combination(s) in the end product evaluation.

2) The tested cells were evaluated for a maximum charge current and maximum voltage limit outlined in the Table below. The end product evaluation shall ensure that current and voltage limits noted are maintained.

This test report shall be read in conjunction with the original report 50183074 001 and 50183074 002.

- Table 1: Electrical parameter of the cell:

Model	Nominal capacity	Nominal voltage	Nominal Charge Current	Nominal discharge current	Max. charge current	Max. discharge current	Upper limit charging voltage	End point voltage
IMR18650-1500mAh	1500mAh	3.7V	300mA	300mA	750mA	750mA	4.2V	3.0V
IMR18650-1200mAh	1200mAh	3.7V	240mA	240mA	600mA	600mA	4.2V	3.0V

Construction:

See original report 50183074 001 & 50183074 002.

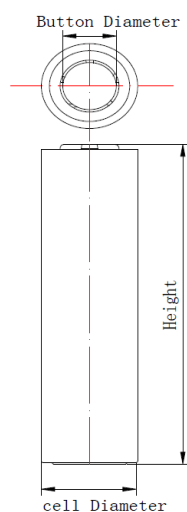
Factory:

Xinxiang Hongli Supply Source Technology Co.,Ltd.

West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China

Description of the product:

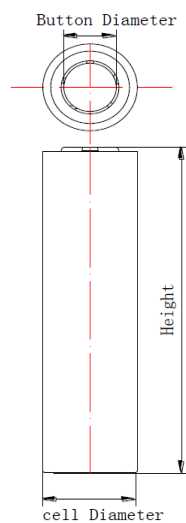
1. Upgraded test standard from "UL 1642: 2020" to "**UL 1642:2020 R10.22**", see Test specification in page 1 for details.
2. Changed discharge temperature range from "-20~60°C" to "**-10~60°C**" for requirement of client.
3. For the client provided error information, corrected cell diameter from



Cell diameter(电池直径) mm	Cell Height(电池高度) mm	Button diameter(盖帽直径) mm
18.0±0.5	65.0±0.3	7.5±0.3

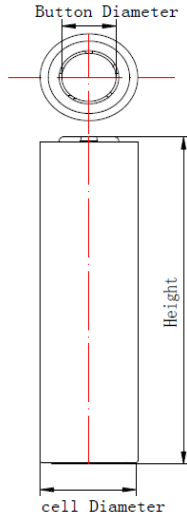
Cell 1 (IMR18650-1500mAh)

to



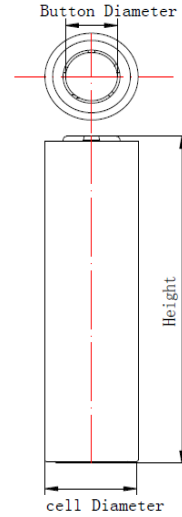
Cell diameter(电池直径) mm	Cell Height(电池高度) mm	Button diameter(盖帽直径) mm
18.0±0.3	65.0±0.5	7.5±0.3

Cell 2 (IMR18650-1200mAh)



Cell diameter(电池直径) mm	Cell Height(电池高度) mm	Button diameter(盖帽直径) mm
18.5±0.3	65.0±0.5	7.5±0.3(平)

Cell 1 (IMR18650-1500mAh)



Cell diameter(电池直径) mm	Cell Height(电池高度) mm	Button diameter(盖帽直径) mm
18.2±0.3	65.0±0.5	7.5±0.3(平)

Cell 2 (IMR18650-1200mAh)

For the above described change(s) the following test item was considered to be necessary:

Change	Testing	Comments	Result
1, 2, 3	N/A	No related test were considered necessary.	P

History of amendments and modifications:

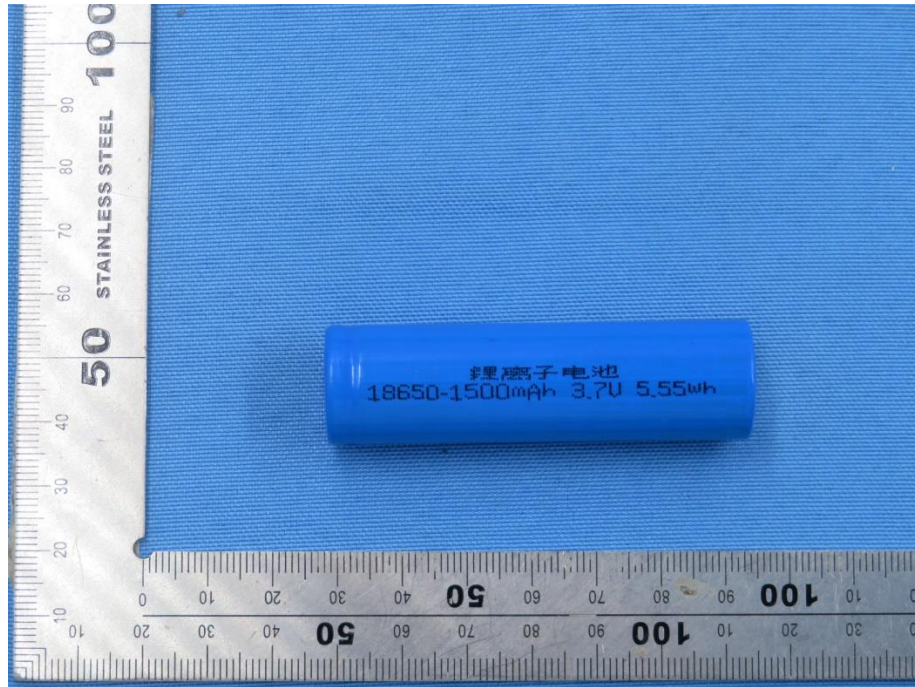
Ref. No. 50183074 001, dated 2018-11-08 (original test report)

Ref. No. 50183074 002, dated 2022-05-11 (1st modification)

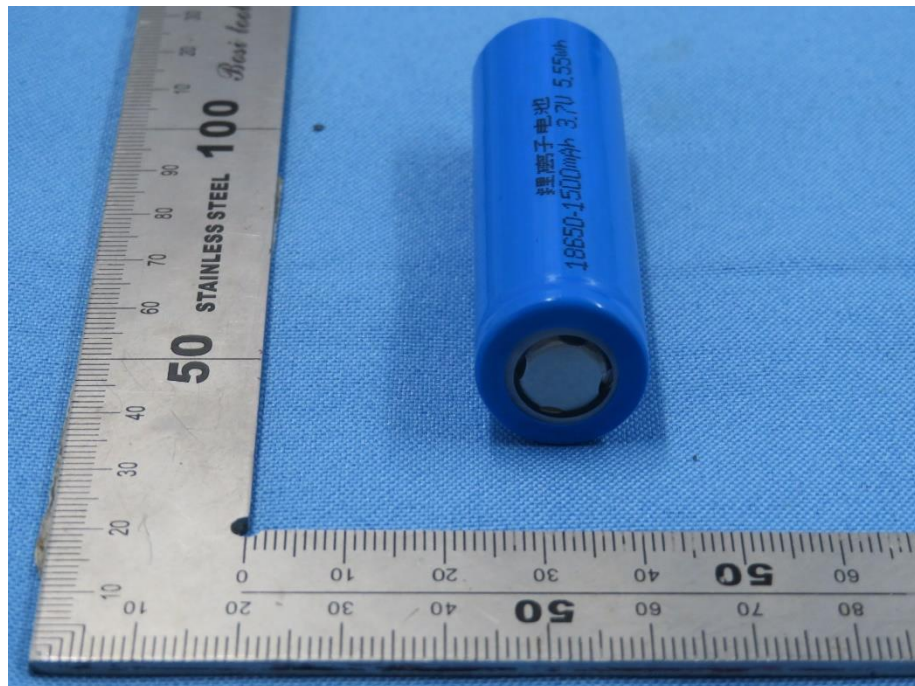
Ref. No. 50183074 003, dated 2024-04-24 (2nd modification)

Product: CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL

Type Designation: IMR18650-1500mAh, IMR18650-1200mAh



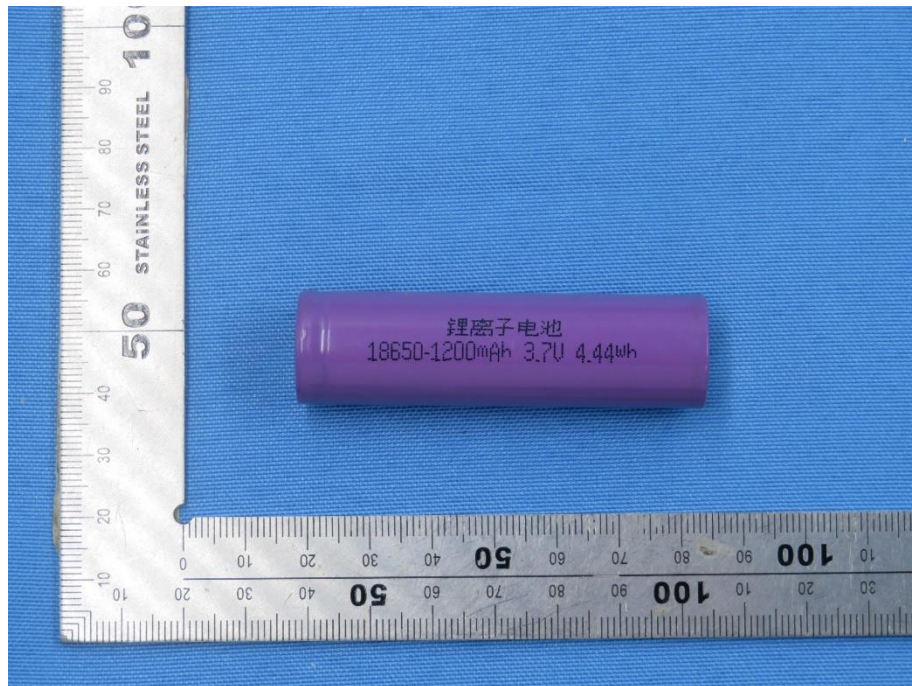
Picture 1. Main view of cell (IMR18650-1500mAh)



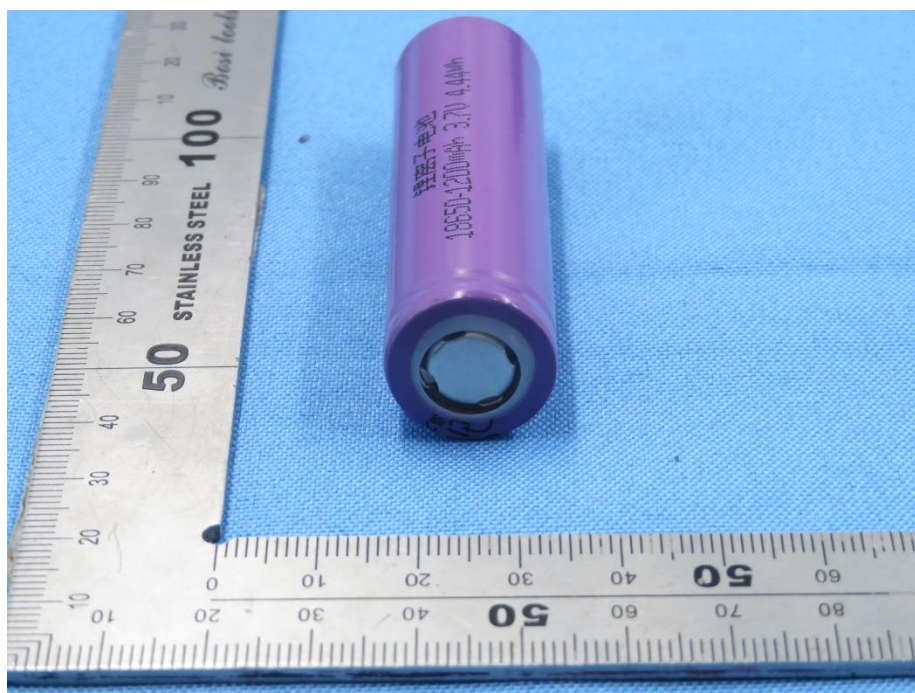
Picture 2. Side view of cell (IMR18650-1500mAh)

Product: CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL

Type Designation: IMR18650-1500mAh, IMR18650-1200mAh



Picture 3. Main view of cell (IMR18650-1200mAh)



Picture 4. Side view of cell (IMR18650-1200mAh)